

LECTURE 25: MATH1061/7861
TOPICS: EQUIVALENCE RELATIONS
DATE: TUESDAY APRIL 28, 2026

Learning objectives:

- (1) Learn the definition of an equivalence relation and equivalence classes.
- (2) Learn how an equivalence relation gives rise to a partition of a set.

Q1: Define ρ on $\mathbb{R}$ by

$$x \rho y \quad \text{iff} \quad x - y \in \mathbb{Q}.$$

(a) Is ρ an equivalence relation? Justify your answer.

✓ Reflexive: $x \rho x$ means $x - x \in \mathbb{Q}$
True because $0 \in \mathbb{Q}$.

✓ Symmetric:

$$x \rho y \rightarrow x - y \in \mathbb{Q} \rightarrow x - y = \frac{a}{b}$$

Transitive: $-(x - y) = \frac{-a}{b} \in \mathbb{Q}$ $a, b \in \mathbb{Z}$
 $b \neq 0$
 $\rightarrow -(x - y) \in \mathbb{Q}$
 $\rightarrow y - x \in \mathbb{Q} \rightarrow y \rho x.$

$$x \rho y \ \& \ y \rho z \rightarrow x - y \in \mathbb{Q} \ \& \ y - z \in \mathbb{Q}$$

sum of rational numbers is rational, $\rightarrow (x - y) + (y - z) \in \mathbb{Q}$

$$\rightarrow x - z \in \mathbb{Q}$$

$$\rightarrow x \rho z.$$

(b) Describe the equivalence class $[1] = \mathbb{Q}$

This is asking: Which numbers are related to 1 under this relation?

$$x \in [1] \text{ means } x - 1 \in \mathbb{Q}$$

This is the same as $x \in \mathbb{Q}$

(c) Is $\underbrace{[\sqrt{2}]}_{\sqrt{2}} = [-\sqrt{2}]$? NO

$$\sqrt{2} \in [-\sqrt{2}]? \text{ i.e.}$$

$$\sqrt{2} - (-\sqrt{2}) \in \mathbb{Q}? \text{ No}$$

Q2: Define a relation ρ on $\mathbb{R}$ by

$$x\rho y \text{ iff } x^2 \leq y^2.$$

Is this an equivalence relation? Justify your answer.

Reflexive: means $x^2 \leq x^2$ ✓

Symmetric: means $x^2 \leq y^2 \rightarrow y^2 \leq x^2$ ✗

Transitive: means $x^2 \leq y^2$ & $y^2 \leq z^2$
 $\rightarrow x^2 \leq z^2$ ✓

Q3: Give some examples of equivalence relations

Set:

Relation

① X

xpy if $x=y$

② X

xpy for $x, y \in X$

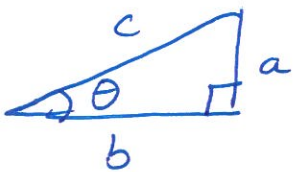
③ $\mathbb{Q}$

xpy if $x+y \in \mathbb{R}$

④ $\mathbb{R}$

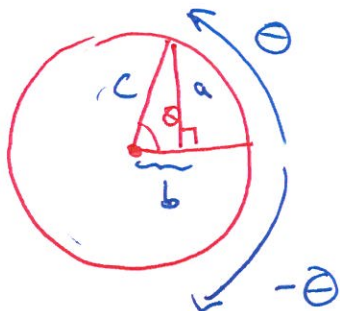
xpy $\cos(x) = \cos(y)$

unnecessary



$$\cos(\theta) = \frac{b}{c}$$

θ is in $(0, \frac{\pi}{2})$



$$\cos(\theta) = \cos(\theta)$$

$$\cos \theta = \frac{b}{c}$$